

Life Processes in Plants

Science — Biology · Class 7 · Max Marks: 17 · Time: 45 min

Section A — Multiple Choice (Higher Order) (1 × 14 = 14)

1. Part of a green leaf is covered with black paper and the plant is left in sunlight. On testing, the covered part shows no starch because:

- A. it had no carbon dioxide
- B. no light reached it, so no photosynthesis
- C. it had no water
- D. it had no chlorophyll

2. A destarched plant is kept in sunlight in a jar containing a chemical that absorbs all the carbon dioxide. Its leaves make no starch because there is no:

- A. carbon dioxide for photosynthesis
- B. oxygen
- C. light
- D. water

3. Which of these is NOT needed for photosynthesis?

- A. water
- B. sunlight
- C. carbon dioxide
- D. oxygen

4. At midnight, a leaf is mainly:

- A. carrying out photosynthesis
- B. doing nothing at all
- C. taking in O₂ and giving out CO₂
- D. taking in CO₂ and giving out O₂

5. Cuscuta is yellow and leafless, so it must live on a host plant because it lacks:

- A. chlorophyll to make its own food
- B. roots
- C. water
- D. stomata

6. Rhizobium bacteria in the root nodules of a pea plant help it by:

- A. providing sunlight
- B. fixing nitrogen for the plant
- C. supplying water
- D. trapping insects

7. A complete ring of bark (with its phloem) is peeled off around a tree trunk. The tree may die because:

- A. it loses all its leaves at once
- B. the food made in the leaves cannot travel down to the roots
- C. water can no longer rise up
- D. it can no longer photosynthesise

8. On a hot, dry, windy day the rate of transpiration in a plant will:

- A. increase
- B. stay the same
- C. fall to zero
- D. decrease

9. A plant lit only by green light grows poorly because chlorophyll:

- A. absorbs green light strongly
- B. mostly reflects green light and absorbs red and blue
- C. needs no light at all
- D. is destroyed by green light

10. The gas bubbles rising from a water plant (Hydrilla) kept in bright sunlight are mainly:

- A. water vapour
- B. nitrogen
- C. oxygen
- D. carbon dioxide

11. Which pair is correctly matched?

- A. phloem – carries water
- B. chlorophyll – transpiration
- C. xylem – carries food
- D. stomata – exchange of gases

12. A mushroom growing on a rotting log obtains its food by:

- A. taking it from a living plant
- B. photosynthesis
- C. absorbing food from the dead log (saprotrophic)
- D. trapping insects

13. If all the stomata of a leaf were sealed with grease, the plant would soon suffer because it could not:

- A. exchange gases or transpire
- B. make chlorophyll
- C. absorb water from the soil
- D. grow new roots

14. Of two lit plants, one has both surfaces of its leaves smeared with vaseline. Compared with the other, this plant will transpire:

- A. only at night
- B. more
- C. exactly the same
- D. much less

Section B — Assertion–Reason (1 × 3 = 3)

Choose: (A) both A and R true and R explains A; (B) both true but R does not explain A; (C) A true, R false; (D) A false, R true.

1. **Assertion (A):** Green plants are called the producers of an ecosystem.

Reason (R): They make their own food by photosynthesis, and other organisms depend on them for food.

2. **Assertion (A):** Water can rise to the leaves at the top of a very tall tree.

Reason (R): The pull created by transpiration from the leaves draws water up the xylem.

3. **Assertion (A):** A pea plant can grow well even in nitrogen-poor soil.

Reason (R): Rhizobium bacteria in its root nodules fix nitrogen for the plant.

Answer Key

Multiple Choice (Higher Order)

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|-------|-------|-------|-------|-------|
| 1. B | 2. A | 3. D | 4. C | 5. A |
| 6. B | 7. B | 8. A | 9. B | 10. C |
| 11. D | 12. C | 13. A | 14. D | |

Assertion–Reason

- 1. A
- 2. A
- 3. A